

Course: B. Tech.

Branch : Electrical

Semester : IV

Subject Code & Name: BTEEPE405B/ Signals and System

Max Marks: 60

Date: 27.08.2022

Duration: 3.45 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

A) Plot the following signals:

Create

6

(i) $e^{-|t|}$

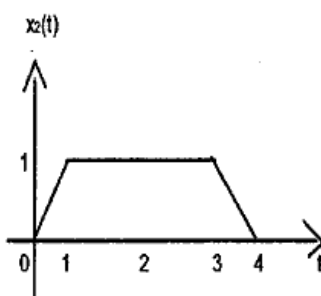
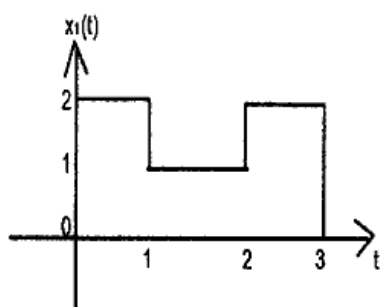
(ii) $r(t) + \delta(t-2)$

(iii) $A\Delta(t/T)$

B) $x_1(t)$ and $x_2(t)$ are shown in following figures:

Evaluate

6



Find:

(i) $x_1(t) + x_2(t)$

(ii) $x_1(t) - x_2(t)$

C) If $X[n] = \{1, 2, 3, 4, 1\}$, then: Find $x[2n-3]$ and plot $x[2n/3]$ and plot $x[-2n]$.

Evaluate

6

Q.2 Solve Any Two of the following.

A) Define systems. Write the properties of the systems.

Understand

6

B) Check the following systems for Static/Dynamic/Causal/Non-Causal

Analyze

6

(i) $y[n] = 2x[n+1] + 3x[n] - x[n-1]$

(ii) $y(t) = \frac{dx(t)}{dx} + x^2(t) + x(t+1)$

C) By using the conventional approach check the causality and time invariant property of the following systems:

Analyze

6

(i) $y(t) = e^{x(t)} + x(t-1)$

(ii) $y(t) = tx(2t) + x(t) + u(t-1)$

Q.3 Solve Any Two of the following.

A) Find the convolution of following signals

Evaluate

6

(i) $x(t) = \text{rect}(t/T)$ and $y(t) = u(t)$

(ii) $x(t) = u(t+5)$ and $y(t) = \delta(t-7)$

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|-----------|---|------------|---|
| B) | How the causality and stability of the system can be checked by impulse response of the system explain in detail. | Understand | 6 |
| C) | The response of discrete time LTI system is $\{2,4\}$ when the input is step. Find the response of the system for the input $\{1,3,3\}$. | Analyze | 6 |

Q.4 Solve Any Two of the following.

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|-----------|---|----------|---|
| A) | Consider a Continuous time LTI system for which input $x(t)$ and output $y(t)$ are related by the following differential equation | Evaluate | 6 |
|-----------|---|----------|---|

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Determine $h(t)$

- (i) When system is Causal
(ii) When the system is stable

- | | | | |
|-----------|---|----------|---|
| B) | Find the Laplace transform of following signals and define ROC. | Evaluate | 6 |
|-----------|---|----------|---|

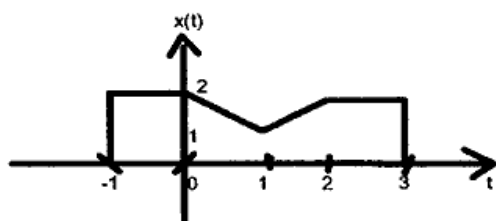
- (i) $x(t) = u(t) \cdot e^{(t+1)}$
(ii) $x(t) = \sin 2t \cdot e^{2t} \cdot u(t)$
(iii) $x(t) = u(t) - u(t-1)$

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|-----------|---|----------|---|
| C) | Find Z- transform of following signals, and comment on ROC. | Evaluate | 6 |
|-----------|---|----------|---|

- (i) $x[n] = \{1,2,3\}$
(ii) $x[n] = 3^n u[n] - 2^n u[n-1]$
(iii) $x[n] = (\frac{1}{2})^n u[n]$

Q.5 Solve Any Two of the following.

- | | | | |
|-----------|-----------------------------|----------|---|
| A) | For $x(t)$ shown in figure: | Evaluate | 6 |
|-----------|-----------------------------|----------|---|



Find:

- (i) $\int_{-\infty}^{\infty} X(\omega) d\omega$
(ii) $\int_{-\infty}^{\infty} X(\omega) e^{j4\omega} d\omega$

- | | | | |
|-----------|--|----------|---|
| B) | Write the properties of Fourier transform. | Remember | 6 |
| C) | Derive the relation between trigonometric, exponential and polar Fourier series coefficient. | Create | 6 |

*** End ***